

HACKCANTON SEASON #2 · GRAND FINAL · 5 AUGUST 2026

tirai.

The confidential multi-dealer RFQ / OTC desk, built native on Canton.
Sealed quotes, atomic delivery-versus-payment, post-trade audit.

You whisper quotes. The market hears nothing.

Asking for a price **is** the information

A bank wants to sell fifty million of bonds. Before it trades, it must ask dealers for a price.

**The moment anyone sees you asking,
they know your size and your direction.**

THE TWO THINGS INSTITUTIONS DO TODAY

Privacy, or proof. Never both.

TRADE IN PUBLIC

Everything leaks

On a transparent chain your request is a transaction and every competing quote is a transaction. Rivals simply read them. Front-running and adverse selection are structural, not incidental.

TRADE ON THE PHONE

Nothing is provable

A voice broker or a chat room is private — and leaves no record. Six months later you cannot prove to compliance that you got the best price available. Nothing is atomic.

**Which is why block trading in 2026
still happens on the telephone.**

TIRAI GIVES YOU BOTH

The dealer terminal, on the ledger

01 · RFQ

Request, unseen

A buyer asks a chosen dealer panel. The market never sees the enquiry.

02 · SEALED QUOTE

Whispered

Each dealer answers the buyer alone. Rivals never receive the contract.

03 · ESCROW

Committed

Quoting locks the dealer's bond. A price is a commitment, not a bluff.

04 · AWARD

Second price

Cheapest ask wins and is paid the runner-up's price. Honest quoting dominates.

05 · SETTLE

Atomic DvP

Bond and cash move in one transaction. The regulator gets the report.

Three execution rails — **reverse-Vickrey auction**, **direct bilateral OTC**, **partial fills** on both. Multi-instrument baskets settle as one atomic multi-leg trade.

You sign in as **one identity** and the home screen is the **book**: every request that party is allowed to see, and nothing else.

NOT HIDDEN BY THE INTERFACE

Dealer B's node **never received** dealer A's quote

On Canton a contract is delivered only to its signatories and observers.
There is no filter to misconfigure and nothing to decrypt – the data never travels.

ENFORCED BY THE LEDGER

Not by policy

A sealed quote is signed dealer + buyer, with no other observers. The regulator observes executed trades only: zero pre-trade visibility, full post-trade record.

FALSIFIABLE, AND FALSIFIED

Live privacy verifier

A script recomputes each party's visible contract set on the real DevNet ledger and exits non-zero if a single quote ever leaks. It is a test, not a dashboard.

THE SUPERPOWER

The chain did the hard part.

TO HIDE ONE NUMBER	WHAT IT COSTS YOU
A trusted execution enclave	Hardware trust, attestation, encrypted handles to manage
Zero-knowledge circuits	Circuits to write and audit, a trusted setup, proving time
Threshold encryption	A committee to run, key shares to hold, liveness to guarantee
Homomorphic encryption	Every branch computed blind, and the performance that implies
Canton	None. Sub-transaction privacy is the ledger model.

"Dealer B cannot see dealer A's quote" is a **signatory / observer** declaration.

WHO BUYS IT

Desks whose tickets are big enough to leak

PRIMARY

Trading desks

Fixed-income and crypto-asset desks at banks, asset managers and prop shops moving \$1m–\$100m tickets that cannot afford to signal.

CHANNEL

Hosting venues

Temple, Bron, Console, Canton Loop — they hold the client relationship and the custody; Tirai ships as an embedded app and shares the fee.

MANDATORY THIRD PARTY

The regulator

Not a payer, but the reason a private chat will not do. Post-trade-only visibility is the middle ground supervisors ask for.

Below a million, leakage is a rounding error.

Above it, leakage is the whole cost.

HOW IT MAKES MONEY

A venue fee the settlement can collect

PER-TRADE FEE

Basis points of notional

Taken in the settlement asset, atomically inside the settlement transaction. If the trade settles, the fee is paid — no invoicing, no collection risk. The rate is not set: it comes out of a design-partner conversation, not out of my head.

NETWORK REWARDS

CIP-0047 activity markers

Featured-app markers accrue on every settlement, so a live desk keeps earning from the volume it clears, on top of the fee.

Fee on notional means one desk clearing institutional size is a real business — and tokenised bond issuance is the growth curve underneath it.

DIFFERENTIATION

Not a DEX. Not a chat room.

ALTERNATIVE	THE SINGLE AXIS IT FAILS ON
Public order book / on-chain RFQ	Pre-trade confidentiality. The order <i>is</i> the signal.
Voice or chat broker	Provable settlement. Nothing is atomic; best execution is reconstructed from chat logs.
Tradeweb / MarketAxess / bank dark pools	The venue itself sees everything. Confidentiality is an operator's promise.
Other Canton privacy demos	No mechanism. They show Canton can keep a secret; they do not run a market.

Private price discovery — **with a receipt.**

LIVE ON CANTON DEVNET – VERIFIABLE RIGHT NOW

Not a mock-up

2

DEVNET PARTICIPANTS
ONE PACKAGE ID

49

SETTLED TRADES
+ 5 ATOMIC BASKETS

16

BEST-EXECUTION
ATTESTATIONS

36

DAML TEST
SCRIPTS GREEN

Deployed on HackCanton's own node and the shared 5N validator under the same package id. Hosted read-only desk over live ledger state at tirai.vercel.app.

Honest label: that ledger history was generated by me. It demonstrates the product works — it is not customer traction, and I will not present it as such.

THE TWO PROOF VIEWS

The claim is checked against the live ledger

VERIFY PRIVACY

Each party's real visible set

Not a claim — a query. Each dealer sees only its own quotes.
The regulator sees zero contracts before settlement. Green, on the network, on demand.

BEST EXECUTION

Proof without an order book

A buyer, or a dealer defending its pricing, discloses a single sealed quote to the regulator. Sixteen attestations prove the clearing price beat every disclosed rival ask.

Confidential pre-trade. Provable post-trade.

YOUR FEEDBACK ASKED FOR REAL EXTERNAL SETTLEMENT

Two assets. Neither of them mine.

The cash leg settles in **real Canton Coin** through the DSO's registry and in **real CBTC** through the DA Utility Registry that BitSafe issues on. Issuers I do not administer and cannot mint into.

CANTON COIN

6 settlements

Four reverse-Vickrey, two direct OTC. 60,900 CC moved to the dealers that won them, each bond-against-cash in one transaction.

CBTC

0.34 + 0.22

A faucet grant accepted on-ledger, then spent through the desk: cleared at the Vickrey second price, and hit directly at the ask.

CETH

One field away

The instrument administrator is the only difference. The rail appears in the desk the day the tokens land, with no release in between.

Nothing in the model is per-asset: `cashInstrument` is any `{admin, id}`, and each registry only differs in where it keeps its off-ledger choice contexts. The desk's Settlement rails view lists every one it can clear on.

THE TEAM

One person. Five days.

WHO

Pugar Huda Mantoro

Final-year undergraduate, building solo. Everything on the previous slides is my own work: the ledger model, the deployer, the desk, the tests, and the two registry integrations.

WHY IT MOVED THIS FAST

Five days, start to deployed

Repo created 22 July. The CIP-0056 settlement leg landed 23 July. Hosted desk live 24 July. Second participant deployed and privacy-verified 26 July. Since the feedback: real Canton Coin and real CBTC settlement on the ledger.

And a written 90-day validation plan with the stop criteria set *in advance*, because a plan that cannot fail is not a plan.

Not a slideshow. The product.

The screenshot displays the 'tirai' application interface, which is a read-only public demo of the Canton Devnet state. The interface is divided into a sidebar on the left and a main content area on the right.

Sidebar (Left):

- Logo: **tirai.**
- Text: CANTON · DEVNET, JSON Ledger API
- Buttons: + New RFQ, Active RFQs, Create RFQ, My activity, Portfolio, Audit trail, Best execution, Verify privacy, Side-by-side proof.
- Navigation: Landing page, Demo video, GitHub, Devnet explorer, Docs.
- Footer: ledger live · pkg 4b1e408f, CANTON IN AS, Buyer — buy side, 10,293,500 USDC.

Main Content Area (Right):

Read-only public demo · live Canton Devnet state — actions are disabled. Clone the repo and run 'npm run demo' to drive it.

You whisper quotes. The market hears nothing. — Landing

Summary Cards:

- 4,695,622** LEDGER OFFSET
- 20** OPEN REQUESTS
- 29** SEALED QUOTES RECEIVED
- 47** YOUR SETTLED TRADES

Active RFQs

every request this party can see — rival dealers see only the ones they were invited to

Filters: All 67, Mine 67, For me 0, Filter by instrument... 67 shown

REQUEST	INSTRUMENT	QUANTITY	MODE	COUNTERPARTY	CASH LEG	PRICE	SEALED QUOTES	STATUS
00eb0578	AAPL30	300	Auction	dealerA + dealerB	USDC	318,000 best	2	Open Settle
00e7d635	BUND10	500	Direct	dealerA	USDC	494,500 best	1	Open Settle
00f27b1e	BUND30	400	Auction	dealerA + dealerB	USDC	1,900,000 best	2	Open Settle
008b4e0e	GILT30	500	Auction	dealerA + dealerB + dealerC	USDC	611,000 best	3	Open Settle
003d748f	IBRD28	1,500	Auction	dealerA + dealerB	USDC	1,487,000 best	2	Open Settle
00fa7bf6	INDON34	800	Auction	dealerB + dealerC	USDC	781,000 best	2	Open Settle
00a9483a	JGB10Y	2,000	Auction	dealerA + dealerB	USDC	1,760,000 best	2	Open Settle
0019b04e	JGB30Y	1,000	Auction	dealerA + dealerB	USDC	2,400,000 best	2	Open Settle
005745bf	MEX34	600	Auction	dealerA + dealerB	USDC	576,000 best	1	Open Settle
006fb2ba	NVDA34	300	Auction	dealerA + dealerB	USDC	341,000 best	1	Open Settle
002e09fe	OAT10Y	600	Direct	dealerB	USDC	580,000 best	1	Open Settle

0:00 / 0:49

A request opened, a dealer's sealed quote, the rival's own session without it, and the two proof views. Recorded against a live ledger, not edited.

TIRAI – INDONESIAN FOR “CURTAIN”

Price discovery happens behind it.

I need one design-partner desk. One hour a week for ninety days – not a purchase order.

1 ONE TRADING DESK AS DESIGN PARTNER

2 CETH AND CBTC TEST-TOKEN GRANTS

3 A HOSTING VENUE FOR A SUPERVISED PILOT